

Assignment - ①

Assessment Sheet

Unit - I : Quantum Mechanics

① An electron is accelerated through a potential difference of 150V. calculate its de-broglie wavelength

potential difference $(V) = 150V$

electron charge $(e) = 1.6 \times 10^{-19} C$

Electron mass $(m) = 9.11 \times 10^{-31} kg$

Kinetic Energy $(K.E) = eV = 1.6 \times 10^{-19} \cdot 150$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2meV}} = \frac{6.625 \times 10^{-34} Js}{\sqrt{2 \cdot 9.11 \times 10^{-31} \cdot 1.6 \times 10^{-19} \cdot 150}}$$

$$\therefore \lambda = 1.00 \times 10^{-10} m = 0.1 nm$$

② An electron is moving with kinetic energy of 5 MeV. find the de-broglie Wavelength Hint: Use $\frac{1}{2}mv^2 = eV$

Given

$$m = 9.11 \times 10^{-31} kg$$

$$e = 5 MeV = 5 \times 10^6 \cdot 1.6 \times 10^{-19} J$$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{6.625 \times 10^{-34}}{\sqrt{2 \cdot 9.11 \times 10^{-31} \cdot 8.0 \times 10^{-13}}}$$

$$\lambda \approx 6.4 \times 10^{-15} m$$

$$\lambda \approx 6.4 fm$$

- ③ A particle is described by the wave function: $\psi(x)$. What is the physical significance of $|\psi(x)|^2$?
- b) sketch the probability density
- c) What is the most probable position of particle
- a) $|\psi(x)|^2 = Ae^{-ax^2}$ represents the probability density of finding the particle at position x .
- b) $|\psi(x)|^2 = A^2 \cdot 29x^2$ is a Gaussian curve centered at $x=0$
- c) Since the peak of the Gaussian is at $x=0$, so the most probable position is $x=0$

- ④ Electrons accelerated through 100V are scattered from a crystal with interatomic spacing $d = 0.1 \text{ nm}$. Find the angle θ at which first order diffraction max using Bragg's law

Accelerating voltage (V) = 100V

Interatomic spacing (d) = 0.1 nm

First-order diffraction: $n = 1$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$V = 100 \text{ V}$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

$$h = 6.625 \times 10^{-34} \text{ Js}$$

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{6.625 \times 10^{-34}}{\sqrt{2 \cdot 9.11 \times 10^{-31} \cdot 1.6 \times 10^{-19} \cdot 100}}$$

$$\lambda = 1.23 \times 10^{-9} \text{ m}$$

- Q) Electrons accelerated through 54V have momentum p.
- calculate the momentum
 - calculate the corresponding kinetic energy and de-broglie wavelength
 - Show how this relates to the observed diffraction at 50°

a) a) $p = \sqrt{2meV}$

$$= \sqrt{2 \cdot 9.1 \times 10^{-31} \cdot 1.6 \times 10^{-19} \cdot 54}$$

$$= 1.57 \times 10^{-24}$$

$$p = 1.25 \times 10^{-24} \text{ kg m/s}$$

b) $KE = eV = 1.6 \times 10^{-19} \cdot 54 = 8.64 \times 10^{-19} \text{ J}$

$$\lambda = \frac{h}{p} = \frac{6.625 \times 10^{-34}}{1.25 \times 10^{-24}}$$

$$\lambda = 5.3 \times 10^{-10} \text{ m}$$

c) Using Bragg's law

$$n\lambda = 2d \sin \theta$$

Assume

$$d = 0.1 \text{ nm} = 10^{-10} \text{ m}$$

$$\lambda = 5.3 \times 10^{-10} \text{ m}$$

$$\sin \theta = \frac{\lambda}{2d}$$

$$= \frac{5.3 \times 10^{-10}}{2 \times 10^{-10}} = 2.65$$

Here $\sin \theta > 1$: which is impossible